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1. Principle of Operation

FMS purpose:

- Improve navigation accuracy
- Aid fuel efficiency
- Reduce crew workload

Two primary guidance modes:

- **LNAV (Lateral Navigation)** — flies aircraft along complex routes on great circle tracks; can fly constant heading, track, or DME arc as required
- **VNAV (Vertical Navigation)** — calculates optimum cruise altitudes; best combination of autothrottle and speed during climb and descent

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2. Control and Display Unit (CDU)

The CDU is the **interface between the aircraft and crew**. Two CDUs are usually fitted either side of the centre console, with the **left CDU normally the master**. On the B747-400 a third CDU is on the centre console for engineering use.

Each CDU has a **monochrome or colour CRT/LCD display** showing "pages" of selected data, plus a selector key panel.

FMC Operating Modes

Mode	Description
Dual Mode (normal)	Both FMCs independently process pilot entries and cross-check results. Same output passed to both FMCs. Provides safety cross-check.
Single Mode	One FMC has failed; the surviving FMC operates the aircraft on its own. Failed FMC may be selected out.
Independent Mode	FMCs decoupled — no safety cross-check between them. Not normal for standard flight profiles.

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3. FMC Data Base

Two major sections:

1. **Performance Data Base:**
   - Aircraft drag and engine characteristics
   - Maximum and minimum speeds
   - Drag and fuel flow factors (maintenance refinable)
   - Replaces need to refer to Cruise Control Manual in flight
   - Provides data for pitch and thrust commands
2. **Navigation Data Base:**
   - Location of navaids, airports, runways
   - SIDs, STARs, approaches, company routes
   - All areas where the aircraft is normally flown
   - Updated by maintenance every **28 days** (matches navigation chart revision cycle)
   - FMC contains **two sets of navigation data**, each valid for 28 days — crew selects which is active during preflight

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4. Initial Actions — FMC Pages (Boeing sequence)

```
graph TD
  A["Power ON → Self-test → IDENT page"] --> B["POS INIT page"]
  B --> C["RTE page"]
  C --> D["PERF INIT page"]
```

Page	Content / Action
IDENT	Confirm aircraft model, operational programme ID, drag/fuel flow factors, Nav DB identifier and cycle. Shows changeover date — if out of date, prompts to change programme.
POS INIT	Check FMS clock against aircraft clock (synchronise for FDR and ETAs). Enter airfield datum or gate position for IRS alignment. Often updated at take-off point for best initial position.
RTE	Enter origin and destination ICAO codes, flight number, and select/enter standard company route.
PERF INIT	Enter fuel weight, fuel reserves, cruise altitude, Cost Index. On B747 — enter fifth engine carriage if applicable.

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5. VNAV — Climb

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Cost Index = 0: Economy speeds = maximum range cruise (minimum trip fuel). VNAV commands:

1. Climb with **climb thrust**, respecting all airspeed and altitude constraints in the SID
2. Climb at economy speed to entered cruise altitude
3. Cruise at economy speed until Top of Descent (TOD)

If flying the climb speed profile would **violate an altitude constraint**, the message **"UNABLE NEXT ALT"** is displayed — pilot must select a different (steeper climb) speed.

Cost Index relationship:
Cost Index = Aircraft operating cost (£/hr) ÷ Fuel cost (£/lb or £/litre)

- **CI = 0:** Minimum trip fuel (max range cruise, low speed descent)
- **Higher CI:** Higher speed → more trip fuel cost but lower time cost → when time is expensive
- **Economy Cruise Mode** gives lowest overall operating cost based on the CI entered

6. LNAV — Cruise

LNAV guidance outputs are normally **great circle tracks** between waypoints. For stored procedures, the FMC can command constant heading, track, or a DME arc as needed.

FMC position determination (in flight):

- Uses inputs from IRS/INS, DME, VOR, and other fitted navigation systems
- FMC present position = **combination (average) of all IRS/INS positions**
- On the ground: FMC uses **IRS/INS only** (no DME/VOR available)
- Requires position from **at least one IRS/INS** to function
- Since inertial systems accumulate errors with time, position error builds up if on ground for extended period
- If significant map error noticed after extended ground delay → **realign IRS/INS and re-enter present position**

LNAV accuracy: Radial error rates of less than **0.05 NM/hour** are not uncommon. GPS input will further improve performance. Operator skill and careful monitoring always remain a deciding factor.

7. Descent

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When a programmed arrival is entered, the FMC calculates a **descent path** based on procedure airspeed/altitude constraints and the **End of Descent (E/D)** — a waypoint with altitude and airspeed constraint coinciding with the final approach fix or runway threshold.

- For **VFR and non-precision approaches**: FMC path built to a point **50 ft over the approach end of the runway**. Crew responsible for not descending below DH without adequate visual contact.
- During missed approach: **LNAV guidance available** to the missed approach point and altitude.

8. CDU Key Groups

CDU keys fall into three major groups:

Group	Function
Alphanumeric Keys	Characters appear in scratch pad for data entry
Function and Mode Keys	System initialization, flight planning, performance selection, active flight plan modification
Line Select Keys (LSK)	12 keys (6 each side of CRT) — transfer data from scratch pad into the selected data field; can also copy displayed data into scratch pad

Key Function and Mode Keys — Summary

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Key	Function
CLB	Current/alternate climb mode — speed/altitude restriction entry, cruise altitude entry
CRZ	Current/alternate cruise mode — optimum altitude, step-climb, turbulence N1 targets
DES	Current/alternate descent mode — target speed, FPA, V/S, V/B
INIT REF	Start-up data pages; IRS initialization; reference data and maintenance pages
N1 LIMIT	Active N1 limit; reduced climb N1 selection (LP turbine RPM)
DEP ARR	Procedure and runway selection at origin/destination
RTE	Flight plan data entry; primary lateral flight plan alteration
LEGS	Detailed data for each flight plan leg — lateral and vertical paths
HOLD	Planning or initiation of holding at a designated waypoint
DIR INTC	Proceed direct to any point, or intercept any leg in the flight plan
FIX	Range and bearing data from present position to an entered fix
PROG	Current flight status — ETA, fuel remaining at waypoint, nav radio tuning, wind, path errors
EXEC	Incorporates displayed data into active flight plan. Illuminated annunciator bar = valid data ready to be made active.
CLR	Clear last character (brief press) or clear entire scratch pad entry (hold >1 sec). Also clears advisory messages.
DEL	Removes data from a display/flight plan after line-selection. Writes "DELETE" in scratch pad, then LSK the item to remove.
+/−	Changes scratch pad data between positive and negative; or inserts a hyphen
NEXT/PREV PAGE	Advances or backs up multi-page displays. Page access wraps around.

9. CDU Illuminated Annunciators

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Annunciator	Colour	Meaning
MSG	White	Alerting or advisory message, or pending messages
FAIL	Amber	FMC failure detected
DISPLAY	White	Page displayed is NOT related to the active flight plan leg or current operational performance mode
OFFSET	White	Parallel offset in use (aircraft flying parallel to, but offset from, the FMS track)

10. Cost Index and Economy

Time Related Cost principle:

- Higher speed → higher trip fuel cost + lower trip time cost
- Lower speed → lower trip fuel cost + higher trip time cost
- **Economy Cruise Mode** finds the optimum balance based on the entered CI

CI modifications possible until passing Top of Descent (TOD): Long Range Cruise (LRC), selected speed cruise.

11. FMS Key Facts Summary

Parameter	Value / Fact
Nav database update cycle	28 days (matches chart revision)
FMC database copies	Two (current and next 28-day cycle)
FMC modes	Dual (normal), Single, Independent
Radial error rate	<0.05 NM/hr typical
VFR/non-precision path end	50 ft over approach end of runway
FMS inputs (navigation)	IRS/INS (minimum 1), DME, VOR, GPS (when fitted)
CI=0	Maximum range cruise (minimum fuel)
UNABLE NEXT ALT	Speed profile would violate altitude constraint in VNAV climb
Waypoint 0 (FMS)	Aircraft's present position (not operator enterable)

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Master exam point: The FMC **cannot detect operator errors** (wrong waypoint lat/long entries). It will navigate precisely to the wrong place if given wrong data. Pre-flight cross-checking of all waypoints against the flight plan is essential.

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